Environment & Sustainable Development

Environment: Sum of all social, economical, biological, physical, and chemical components constituting man's surroundings.

Components of Environment

- **Biotic Components:** All living elements (flora, fauna, man).
- **Abiotic Components:** Non-living elements (lithosphere, hydrosphere, atmosphere).
- **Energy:** Solar, geochemical, thermoelectric, hydroelectric, nuclear atomic energy.

Functions of the Environment

1. **Provides Resources for Production:** Supplies raw materials (soil, water, forests, minerals, marine life).
   - **Renewable Resources:** Can be used without depletion (trees, animals).
   - **Non-renewable Resources:** Get exhausted with use (fossil fuels like natural gas, coal, oil).
2. **Sustains Life:** Includes sun, soil, water, and air, which are necessities for all living things.
3. **Assimilates Waste:** Absorbs waste generated by production and consumption activities.
4. **Enhances Quality of Life:** Natural elements like land, forests, oceans, mountains, and deserts offer beauty and enjoyment.

Carrying Capacity: The maximum limit up to which an ecosystem can support the existence of a population. It includes the environment's capacity to provide resources and absorb degradation.

Environmental Problems

1. Pollution

Activities of production and consumption that challenge the purity of air and water.

Air Pollution
Mixture of solid particles and gases in the air.

- **Causes:**
  - Emission of gases from motor vehicles.
  - Smoke from industries (especially coal-based).
  - Agricultural activities (insecticides, pesticides, fertilizers).
  - Mining operations (dust and chemicals).
  - Indoor air pollution (cleaning products, paints).
- **Control Measures:**
  - Promote public transport and carpooling.
  - Use cleaner fuels (electric cars, CNG).
  - Conserve electricity (switch off lights/fans).
  - Utilize clean energy technologies (solar energy).
  - Use cleaner household fuels (LPG).

Water Pollution
Presence of toxic chemicals in water, harmful to health and environment.

- **Causes:**
  - Domestic sewage into rivers.
  - Industrial waste discharge.
  - Insecticides and pesticides run-off.
  - Washing vegetables in untreated sewage water.
- **Consequences:** Hepatitis, Typhoid, Diarrhoea, Cholera.

Noise Pollution
Harmful impact of excessive noise on human or animal life.

- **Sources:** Machines, transport, vehicles with loud horns.
- **Consequences:** Irritation, hypertension, hearing loss, sleep disturbances.

2. Excessive Exploitation of Natural Resources

Natural capital (forests, minerals, soil) is depleted without adequate provision for replenishment, unlike physical capital.

Deforestation
Permanent removal of trees for agriculture, timber, construction.

- **Consequences:** Land degradation, biodiversity loss, ecological imbalance, air pollution.

Land Degradation
Decline in quality of soil, water, or vegetation due to human activities.

- **Causes:**
  - Soil erosion (wind, floods).
  - Overgrazing.
  - Excessive use of fertilizers and pesticides.
  - Extraction of excess groundwater.
  - Improper crop rotation.

Biodiversity Loss
Disappearance or reduction of plant and animal species.

- **Causes:**
  - Pollution.
  - Greenhouse effect (rising temperatures).
  - Population growth and overconsumption.

Global Warming
Gradual increase in Earth's average temperature due to increased greenhouse gases (primarily CO_2, methane).

- **Consequences:**
  - Increased average temperature.
  - Rising sea levels.
  - Modified rainfall patterns.
  - Natural calamities (floods, famines, heatwaves).
  - Glacier melting.
  - Emergence of new diseases.
  - Effect on marine life.

Chipko Movement
Social movement using non-violent resistance (hugging trees) to protect them from felling. Started in Mandal, Uttarakhand in 1973 by Sunderlal Bahuguna. Similar movement "Appiko" in Karnataka.

Pollution Control Boards

Central Pollution Control Board (CPCB) established in 1974 to address water and air pollution.

- **Functions:**
  - Promote cleanliness of streams and wells.
  - Improve air quality and control air pollution.
  - Organize awareness programs.
  - Collect and publish technical and statistical data.

Sustainable Development

Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Sustainable Development Goals (SDGs)

17 global goals set by the United Nations in 2015, to be achieved by 2030 (e.g., no poverty, zero hunger, quality education, climate action).

Aims of Sustainable Development

- Sustainable and equitable use of resources without environmental damage.
- Prevent damage to life-support systems.
- Conserve biodiversity and resources for long-term food security.

Strategies for Sustainable Development

1. **Use of Non-conventional Energy Sources:**
   - Shift from polluting conventional fuels (firewood, fossil fuels) to renewable and eco-friendly sources like wind, solar, tidal, geothermal, and biomass energy.
2. **Use of Cleaner Fuels:**
   - Promote LPG and Gobar gas in rural areas (Ujjwala scheme).
   - Promote CNG in urban areas for vehicles (lower emissions).
3. **Shift to Organic Farming:**
   - Avoid chemical fertilizers, pesticides, and insecticides.
   - Rely on crop rotation, animal manures to improve soil health and prevent water pollution.
4. **Establishment of Mini-hydel Plants:**
   - Utilize stream energy in mountainous regions to generate local electricity, avoiding large transmission infrastructure.
5. **Wind Power:**
   - Install wind turbines in high-wind areas to generate clean electricity, despite high initial costs.
6. **Traditional Knowledge and Practices:**
   - Integrate traditional knowledge (e.g., plant-